

# StoryPrompt: Exploring the Design Space of an AI-Empowered Creative Storytelling System for Elementary Children

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## ABSTRACT

Creative storytelling is a crucial learning activity for children. We conducted formative research with three teachers and 18 children and derived five design implications. Based on the findings, we developed StoryPrompt, an interactive system that enables elementary children to co-create stories and comics with generative AI, aiming to enhance their literacy and creativity. The preliminary evaluation, involving eight children and three HCI experts, demonstrated good system usability and children's satisfactory learning experience. We discuss the advantages and considerations of our design features in using generative AI to empower creative storytelling for elementary children.

## CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interactive systems and tools.

## KEYWORDS

Generative AI, Storytelling, Children, Creativity, Literacy, Design

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## 1 INTRODUCTION

Creative storytelling, which involves vocalization, narrative structure, and mental imagery, is essential for enhancing children's various skills such as literacy, problem-solving, and creativity. Recognizing its significance, HCI research has extensively explored the design space of various tools to support children's storytelling, ranging from interactive e-books [9, 18], computer games [3], augmented/tangible toys [11, 31], to embodied systems [10]. These tools enable children to participate in the story creation process to a certain extent by employing various multimedia narrative elements (e.g., characters, settings), and most emphasizing collaborative storytelling practices.

Recent HCI work has turned towards artificial intelligence (AI) to create dynamic, personalized, and effective storytelling tools [4, 13, 29, 30]. Differing from traditional systems, AI-driven tools offer novel means of generating content from user inputs [13, 29]. While existing AI storytelling systems for children focus on visual storytelling using generative AI tools (e.g., Stable Diffusion, SD) [4, 13, 29], this approach can limit narrative complexity, especially for older children who require more sophisticated storytelling practice [23]. More critically, there is also a gap in understanding if and how generative AI can enhance story quality and creativity. Therefore, we aim to explore the design space of an AI-empowered interactive system to support children's high-quality and creative storytelling practices.

Based on the formative research, we developed StoryPrompt, an interactive system that enables elementary children to co-create stories and comics with generative AI (ChatGPT and SD), aiming to enrich their literacy and creativity. Unlike previous AI visual storytelling tools [4, 13, 29], our system (1) incorporates both *story*

*structure planning* and *emergent storytelling* which enable children to create complex narratives; (2) leverages large language model (LLM) and AI-generated texts and images to provoke creativity through the use of words, problem-solving, and visual inspiration and engagement. Our preliminary findings suggest good system usability and a satisfactory learning experience of children. We discuss the considerations and advantages of designing AI-empowered creative storytelling systems for elementary children.

This study contributes to the child-centric AI design in two ways. *First*, our research prototype leverages generative AI to support children's storytelling for complex narration. *Second*, the design rationale and implications derived from the findings can benefit future research in creating AI-empowered creative storytelling systems for children.

## 2 BACKGROUND

### 2.1 Children's Storytelling Practice

Research shows that children as young as two begin to understand *story structures* (e.g., "Once upon a time" as a formal opening [26]), while by age four, they can grasp story elements like *characters*, *settings* and *plots* [12]. Beyond these, the *character* [1, 5], *emotion* [19, 23], *prop* [13, 14], and *action* [13, 14] are also key, with the *character (traits and emotions)* being particularly influential on the flow of plots and narratives [1, 5]. Visual character representations can further affect story quality and creativity [16, 27]. Research also suggests that children have an innate love of stories and fairy tales and prefer to use their own life stories as storytelling materials [27]. The inclusion of nonsensical and fantasy-based narrative elements in stories not only engages children but also inspires their creativity and imagination [3].

Non-digital tools, activities, and strategies are used to support children's storytelling [7, 28, 30]. Physical toys, visual pictures, drawing activities, and embodied enactment (e.g., fantasy play) are effective in engaging children in visual thinking and embodied storytelling experiences [7]. Another frequently used strategy is *guided conversation*, where a storyteller asks children questions about story content and provides responsive feedback [28, 30]. This strategy highlights the importance of involving reading experts (e.g., teachers, system guided instructions) to provide scaffold. Many studies have also demonstrated the effectiveness of storytelling practices in enhancing a wide range of skills in children, such as *creativity* (e.g., problem-solving skills, imagination, creative storytelling) [3, 27, 29], *literacy* (e.g., narrations, languages, writing skills) [17, 26, 30], and *social-emotional development* (e.g., empathy) [27, 30]. Other associated benefits include increased learning interests, motivation, and memorization [22, 29, 30]. In this study, we focus on **literacy (story quality)** and **creativity (story creativity)** as these are particularly important for elementary children in various grade levels.

### 2.2 Interactive and AI-based Storytelling Systems for Children

Digital storytelling systems for children employ various models such as *emergent storytelling*<sup>1</sup> [18, 21, 25], *character-driven storytelling*<sup>2</sup> [5, 6, 16], and *user-centered plot resolution*<sup>3</sup> [8]. For example, Graphic Storywriter is an interactive computer software that supports children's emergent storytelling [26]. It allows the child to select story scenes, characters and objects, after which the associated texts are generated. Story Creation is a character-driven storytelling system that allows children to use an avatar to enact a story [27]. The Interactive Storytelling Robot is a tangible system that allows children to make decisions on several narrative branches through dialogues with a robot [15].

AI techniques are increasingly used to support collaborative storytelling, predominantly visual. Bernal et al. designed Paper Dreams, a web-based platform that uses AI to convert users' text into images [4]. Han and Cai presented Alstory, a web-based app that allows children to compose visual story with AI [13]. This app uses pre-defined stickers like characters and facial expressions, and the AI generator to create story images. Zhang et al. presented StoryDrawer, which turns children's live narration into AI-generated drawings [29]. The results showed that StoryDrawer provoked children's creative and elaborate ideas, contributing to their creative outcomes during an engaging visual storytelling experience. However, the three designs heavily rely on children's visual thinking ability (drawing all the elements in a single page), which may limit complex sequential storytelling. Another system presented by Zhang et al. called StoryBuddy, enables parents to collaborate with AI in creating question-answer pairs for children [30]. This project aims to use AI to generate questions and facilitates guided conversation during parent-child storytelling processes.

In summary, generative AI storytelling tools that support elementary children in co-creating high-quality and creative stories are underexplored. Therefore, our research question is: **How can we design a generative AI storytelling tool that effectively aids elementary children in co-creating high-quality and creative stories, potentially enhancing their literacy and creativity?**

## 3 FORMATIVE RESEARCH

We performed formative research involving three elementary teachers and 18 children to understand current storytelling educational practices and derive design implications. The study was approved by the first author's university's ethical review board.

### 3.1 Methods

We conducted in-depth interviews with three female teachers, and each had ten years of teaching experience in the subject of Chinese. The interview aimed to understand current storytelling instruction in schools, including learning goals, effective training techniques, and potential challenges. Each interview lasted 30-60 minutes and took place in a quiet classroom. Two researchers participated,

<sup>1</sup>The user creates a story by selecting certain narrative elements (characters and backgrounds) from a large corpus of materials such as [25].

<sup>2</sup>The user plays as one of the characters and enacts the story [5].

<sup>3</sup>The user participates in the storyline and dynamically influences the development of the plot through decision-making process [24].

with one primarily responsible for asking questions and the other for taking notes based on our study protocol (Appendix A). The interviews were audio recorded, and each participant received \$30 as research compensation. We also conducted three rounds of focus groups with 18 children, involving 9 boys and 9 girls from Grade 2 to Grade 4, with an average age of  $8.00 \pm 0.84$ . The focus groups aimed to understand children's prior experiences with computers and AI, their storytelling strategies, and their preferences for story characters and settings (Appendix B). Each focus group contained six children and two researchers, with one researcher serving as the moderator (asking questions) and the other as the facilitator (taking notes). Each round lasted 30–40 minutes, taking place in a quiet classroom. The focus groups were recorded. Each child received a maker suite as the gift. The audio recordings of the focus groups were transcribed into text. Three researchers used thematic analysis to analyze the qualitative data. We obtained consent from the teachers and the children (and their guardians).

### 3.2 Key Findings

All teachers agreed that creative storytelling was significantly critical for children, aiding the development of strong creative writing skills essential in elementary education. They highlighted several points: **(1) Learning goals:** Younger children (Grades 1–2) focused on learning literacy words, retelling techniques, and crafting simple stories from personal experiences. In contrast, older children (Grades 3–6) aimed to create stories with complex structures. **(2) Effective training techniques:** Guided instructions, such as asking *purposeful questions* (e.g., “If you were the character, what would you do?”), *offering literacy prompts* (e.g., offering hints for words or phrases), *providing visual aids* (e.g., providing concrete pictures to describe), and *relating to other references and materials* (e.g., relating to books the children had read before), supported learning. A general story structure plan coupled with room for emergent storytelling and iterative refinement was key for quality narratives. Multimedia materials, such as images, also enhanced children's learning interests. **(3) Potential challenges:** Limited resources beyond books challenged the development of creative storytelling and writing. Digital storytelling tools suitable various grades were scarce. **(4) Other feedback:** Schools possessed digital tools like computers and iPads, and children of that age were familiar with computer functions.

All children reported prior computer and iPad app experience, but only four children had used AI-related products. Many children reported that, encouraged by their teachers, they would use themselves as the reference for a story character, making the story vivid and realistic. We identified four types for children's preferred characters: *talking animals or objects* (e.g. talking cats), *people with magic or superpowers* (e.g., witches), *sci-fi people* (e.g., scientists), and *normal girls or boys*. The four preferred scenes included: *natural scenes* (e.g., forests), *fantasy scenes* (e.g., magic schools), *sci-fi scenes* (e.g., labs), and *everyday scenes* (e.g., restaurants). We also noticed that higher-grade children preferred to choose sci-fi and fantasy-related scenes and characters while lower-grade children liked objects or animals or themselves. In terms of gender differences, boys showed more preferences toward sci-fi scenes and characters compared to that of girls. These findings helped us to

identify five design implications and provided useful examples (e.g., characters, settings) for the system design.

### 3.3 Design Implications

Based on the empirical results, we have identified five design implications. **DI#1:** The system should support varying difficulty levels of creative storytelling activities for children across different grades. **DI#2:** The system should allow children to plan the story structure while also opening possibilities for emergent storytelling, serendipitous discovery, and iterative creation. **DI#3:** Generative AI could be used to provide both verbal (words) and nonverbal (visual images) prompts to inspire children's creative storytelling and learning interests. **DI#4:** Purposeful guidance is important in children's storytelling. **DI#5:** Story characters and settings should be designed based on story genres and children's preferences.

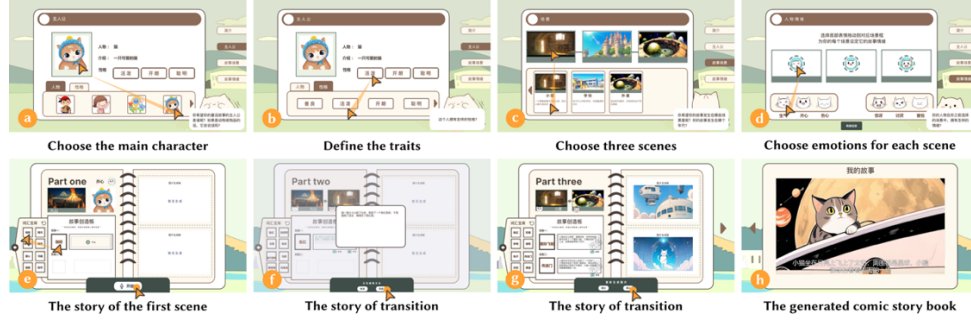
## 4 THE AI STORYTELLING SYSTEM DESIGN

### 4.1 Prototype Overview

StoryPrompt is an interactive system running on a computer that enables elementary children (ages 7–12) to co-create creative stories and comics with generative AI. The system can be used by an individual child or by a group of 2–3 children. The system is especially recommended for use with a teacher for children below Grade 3. While interacting with StoryPrompt, the child begins by selecting a character (Figure 1a) and defining the character's traits such as bravery and honesty (Figure 1b). Next, the child selects three sequential story scenes (Figure 1c) and defines the character emotions associated with each scene (Figure 1d). Once the story setup is complete, the main task is to fill the story gap to make a logical, believable, and coherent story [20]; and the process requires the child to relate AI-generated keywords to the other story elements such as character emotions and story scenes (Figure 1e–f). Specifically, the child needs to select the keywords (nouns) from a set of words generated by ChatGPT to create two storylines (two paragraphs) for each scene (Figure 1e). The storylines could describe how the character and the objects (keywords) come to the scene and why the character experiences such emotions with the objects. The child can record his or her voice to generate the text and then modify the details using text input (Figure 1f). The generative keywords are updated in real-time based on the story setup and the child's story input. After each storyline is created, the system generates a comic-styled image using our trained SD model (Figure 1g). The child needs to complete six storylines for three sequential scenes to finish the entire story. At the end, the child can view and download the generated comic storybook consisting of six pages (Figure 1h).

### 4.2 Design Features

**4.2.1 Story Characters, Settings, and Emotions.** Children first *plan the story structure* by selecting a main character and his/her three associated traits, and the three settings the character will be in, as well as his/her emotion in each setting (DI#2, Figure 1a–d). Although previous work suggests that common characters in fables may have their associated traits (e.g., the prince is brave) [1], the system enables to redefine the character traits (could be these with



**Figure 1:** (a) Selecting the main character by dragging the image into the profile box; (b) defining the character traits by dragging three labels into the boxes; (c) selecting three scenes by dragging the images into the boxes; (d) defining the three associated emotions for each scene by dragging the emotion icons into the boxes; (e) selecting the AI-generated keywords by dragging them into the boxes and using the keywords to create storylines for the 1st scene; (f) using voice recording to generate storylines and modifying them using text input for the 2nd scene; (g) generating AI images for the 3rd scene; (h) viewing the entire generated storybook.

the consistent or conflicting ones to the public common impression), thereby fostering more creative ideas. We incorporated four types of ten characters and four types of seven settings based on the formative study (DI#5). Various settings and emotions are used to trigger the story transition because changes in character locations or emotions often predict the story development and can serve as a good starting point to expand imagination. And emphasizing on emotions may also benefit children’s emotional intelligence [2], the ability of which is important to emphasize with imaginative characters during the storytelling. The system is flexible in supporting the selection of the same or different settings or emotions, enabling simple or complex story generation (DI#1).

**4.2.2 AI Generative Keywords as Narrative Prompts.** In each setting, children can view the eight keywords generated from ChatGPT (DI#3, Figure 1e). Children’s main task is to drag the selected keyword into the box and create the storyline by relating this keyword to the story elements (DI#2). If they are unsatisfied, they can update the keywords. The initial eight keywords are generated based on the selected character (traits), settings, and emotions, while the updated ones are generated in real-time from the children’s full text input. Among the words, four displayed in the first two lines are strongly associated to the storyline (e.g., generating the words “queen” and “crown” when a child says, “A princess is reading in the garden.”) while the other four shown in the last two lines are not (e.g., providing the words “moonlight” and “treasure” when a child says, “A cat walks in the forest.”). We did not tell children the differences between these words to avoid potential impacts on their learning confidence. This design accommodates the varying difficulty levels of the storytelling tasks (DI#1).

**4.2.3 AI Generative Visual Images.** Children create the entire story by completing six story paragraphs (storylines) one by one (2 paragraphs per setting  $\times$  3 settings) (Figure 1f). Once the children are satisfied with each storyline, they can create its image (DI#3) (Figure 1g-h). They can regenerate the image until satisfied.

**4.2.4 Others.** The system incorporates a set of questions based on teachers’ interviews to provoke meaningful dialogues (DI#4).

Several questions include “How did the character arrive at this scene? Why does the character have such an emotion?”. Teachers can read the questions aloud to younger children while older children can use them for self-guidance. Story input can be done through voice recording or text input, making it accessible for children of different ages (DI#1).

### 4.3 Technical Implementation

The system was implemented using Unity. For real-time AI keyword generation, we employed prompt engineering and provided cases to control ChatGPT (e.g., “Output eight Chinese words based on the characters, scenes, and keywords I entered, the first four of which should be strongly correlated with my input”), enabling us to generate four strongly and four weakly associated keywords from user input text that are appropriate for children. For real-time AI image generation, our process involved calling the ChatGPT API to extract key characters, scenes, and actions from the corresponding text and adding details (e.g., “Extract key elements from my input, and these elements should include character, scene, and behavior”, “Describe those elements and add details in three or four phrases”, “Output according to the examples”). We utilized the SD API to generate images based on the extracted information. To achieve aesthetic cartoon-style images appealing to children, we trained our SD model. We used a formula consisting of Midjourney theme keywords, lens, style references, adjectives, and suffix words to generate the SD training materials and maintain visual style consistency. Voice recognition and audio-text translation were implemented using the Baidu API.

## 5 PRELIMINARY EVALUATION

Before examining the learning effectiveness, we conducted a preliminary study with eight children and three HCI experts to assess the system’s usability and users’ learning experiences.

### 5.1 Methods

Eight children (4 boys and 4 girls, from Grade 1 to Grade 5, with an average age of  $8.63 \pm 2.13$  years) who did not participate in our



**Figure 2: (a) The system being played by an individual child (Jessica); (b) the system being used by children in pairs (Lisa and Robert); (c) the part of the storybooks created by an individual child (Jessica); (d) the storybooks created by children in pairs (Lisa and Robert).**

formative study took part in this evaluation. All the children had prior experience playing computer or iPad apps, but only three had explored AI products. Four children used our system individually while the other four used it in pairs (Figure 2). Both conditions were facilitated by a trained researcher. The researcher first briefly introduced the study goal and demonstrated the system to the children. After that, the children used the system to create a story. Afterward, they were asked to rate the quality of the system design, the created story, and the generated comic books on a scale from 1 (lowest) to 10 (highest). They were also asked about their likes and dislikes regarding the system and whether they would like to continue using it in the future. The children’s demographic information was also collected (Appendix C). Each child received a maker suite as a gift after the study. The study was conducted in a quiet classroom, lasting around 20 to 40 minutes for each child/pair group. Three researchers participated, with one facilitating children’s use while the other two setting up the equipment and taking observational notes. We also invited three HCI experts to review our system. There were two females and one male, with over ten years of experience in child-centric and AI design. We first introduced our design goals and then demonstrated the system to the experts, followed by soliciting their feedback. Two researchers took notes on the feedback provided. The entire process took place in our lab and lasted around one hour. Three researchers used a thematic approach to analyze qualitative observations and interviews and used descriptive analysis to analyze children’s stories and rating scores.

## 5.2 Results

**5.2.1 Children’s Use of the System.** We observed that all the children (even the younger ones) could quickly learn to use the system only after one demonstration. All the children stayed focused during use, whether they were interacting the system alone or in pairs. They successfully created six stories with 1830 Chinese words and 36 images (Appendix C). The six stories contained 60 sentences, involving 21 short ones with simple grammar structures and 39 long ones with complex structures, with an average of 10 sentences (avg 3.5 short sentences and avg 6.5 long sentences) in each story. As for genres, the children created one dream narrative story, two fantasy stories, and three stories with combined genres involving sci-fi + fiction, sci-fi + adventure, and fantasy + adventure. Many examples of children creatively using AI-generated keywords to develop and modify their storylines or enrich the story details were noted. For example, P3 introduced the new character “elf” and detailed the story happened with *elf* and the main character “witch”.

P1 ideated a bubble-shaped aircraft by using the word “bubble”; by saying, “a cluster of bubbles enveloped the kitten and transported it to outer space”, P1 naturally transitioned from the *forest* setting to the *outer space* setting.

We also noted several differences in children’s behaviors across ages or under various use contexts. When the children were asked to use keywords for story creation, most younger children did not hesitate to pick words, while older children hesitated more and tried updating the words several times to look for the exact ones they were expecting. This may be because younger children did not have too many thoughts initially, while older children already have some plans. However, once the facilitator used the guided dialogue to encourage the older children to try the words and rethink more possibilities for the storyline (e.g., “Think about how the character got the object and arrived at the scene. Try to describe the process within 1-2 simple sentences”), they started to carefully select keywords and use them to ideate their stories. We noticed that older children with high literacy skills tended to choose weakly associated keywords, while younger children tended to choose strongly associated keywords. All the children became extremely excited once they saw the generated images (e.g., stared at the screen and saying “Wow!”), even though the images sometimes did not match their expectations. We also observed that younger children enjoyed using voice interaction while older children preferred typing text directly. Additionally, younger children working in pairs intended to use turn-taking for decision-making while older children preferred to negotiate and resolve the conflicts together.

In the post-interviews, the children rated the system design quality as  $8.75 \pm 1.28$  (out of 10), the created story quality as  $8.75 \pm 0.88$ , and the generated image quality as  $7.75 \pm 2.05$ . All the children expressed a strong willingness to continue using the system in the future (e.g., “Definitely I want to use it”). When asked about their favorite system function, most children mentioned the AI-generated images (4/7), followed by the character selection (3/7), and AI-generated keywords (2/7). As explained by Jessica (pseudonym), “It’s so magical that the system can generate images exactly as I imagined just by clicking a button” while Michael stated, “The generated images provided me with new inspiration for my story”. Alice mentioned, “I am glad to be able to freely choose the character and define their traits...The generated keywords provided a new way of thinking about story creation that I had never considered before. I would like to use it in my future creative writing exercises”. The results were consistent with our observations that many children initially found using these keywords challenging, but once they tried and succeeded, they become more confident. We also found

most children expressed a great sense of accomplishment and satisfaction in being able to use these nonsensical keywords to create a coherent story when asked to rate their own stories. Interestingly, when asked for suggestions for the system, most children mentioned the same three preferred functions mentioned earlier. Four children mentioned that the AI-generated images differed from their expectations (e.g., containing a smaller size of the character, lack of coherence), and three suggested adding more options for characters and settings, while two suggested increasing the options for keywords. The study showed the effectiveness of the generated words in triggering children’s problem-solving and reflective moments and revealed the limitations of the generated images in accurately representing children’s storytelling ideas, despite their impacts on increasing children’s learning interests and sometimes invoking inspirations.

**5.2.2 Experts’ Feedback.** The three experts generally found the design interesting and believed it could benefit children in creative storytelling and writing. They also pointed out several issues with the current design, including the need to add audio instructions to guide children’s interaction flow and add sound effects to further engage children. Two experts who specialize in designing learning interfaces for children suggested adding a cartoon-style virtual tutor to guide children’s use and provide instructions on their storytelling.

## 6 DISCUSSION, CONCLUSION, AND FUTURE WORK

We present design implications, a design example (StoryPrompt), and a preliminary evaluation of the AI-empowered creative storytelling system for elementary children. We discuss three advantages and considerations.

First, unlike prior generative AI storytelling systems that focus on either visual storytelling [4, 13, 29] or generating purposeful questions for parents [30], we leverage generative AI to provide both verbal (words) and nonverbal (visual images) prompts to promote children’s creative storytelling. When children use StoryPrompt, we observed that the AI-generated words provided younger children some ideation resources to start their stories while offering older children challenging tasks for them to reflect, rethink more possibilities, and potentially change the original direction, both of which can benefit children’s creativity. We also noticed that the generated images mainly engaged children rather than inspiring them, given the limited and inaccurate AI visual generation capability. This may suggest the ineffectiveness of purely including AI-generated images in supporting children’s creative storytelling.

Second, it is crucial to balance “overall planning” and “emergent storytelling” for high-quality creative story creation. A good story usually contains a mature and potentially complex narrative structure that is planned in advance by the storyteller [26]. Therefore, it is beneficial to guide children in planning the global story structure first and then enable them to leverage the advancements of AI to fill the story gaps with serendipitous details, while maintaining their creative ownership during the child-AI collaboration [4]. In the study, the children were allowed to select their preferred keywords and regenerate keywords or images. Finally, all the children successfully completed the tasks, generating stories with various

genres (including three complex combined genres) and more complex sentences than simple ones. They also gave high scores for their story quality, suggesting the effectiveness of our design. However, we acknowledged that the story quality and creativity were not formally evaluated by teachers, which may require further research.

Third, to accommodate children with varying literacy skills, it is important to maintain design flexibility for contextual use. Taking StoryPrompt as an example, teachers can adjust task difficulty by changing the setup of the character, settings, and emotions (e.g., a story with a “happy-sad-happy” emotional trajectory would be easier compared to an “angry-sad-surprise” one). Encouraging the use of strongly or weakly associated words is another strategy. In the evaluation, we indeed noticed that older children with higher literacy skills appeared to select more weakly associated keywords. Although we have used prompts to ensure the suitability of the keywords generated by ChatGPT for children, due to the untransparent and uncontrollable nature of AI, it is also important to involve adult experts in its contextual use for ethical considerations.

We acknowledged several limitations of our system design and study design. First, we did not recommend the story length for children in different grades, which may have resulted in variations in story length and complexity. Second, as a preliminary evaluation, we only had a small sample size and primarily focused on qualitative results on children’s system use and learning behaviors. We neither rigorously assessed the levels of creativity and literacy in children’s generated stories, nor invited educators to review the system effectiveness. Third, we incorporated the group-based and individual uses to demonstrate the system feasibility in various use contexts, but the analysis of children’s dynamic social relationships and its impact on their collaboration were not considered and beyond the current research scope.

In the future, we plan to revise the system and conduct a formal evaluation with elementary children and teachers to examine the effectiveness of the system in supporting children’s literacy and creativity. Our goal is to investigate whether children can produce high-quality creative stories using the system compared to traditional instruction in schools. Additionally, we aim to identify design features that may benefit their development, with the hope of contributing to the design knowledge in child-centric AI design for creative storytelling tasks.

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## APPENDICES

### A INTERVIEW PROTOCOL WITH TEACHERS

1. Do elementary children receive creative storytelling training at schools? If yes, when do they start? What are the learning goals (e.g., language, creativity, problem solving)? Do the learning goals vary for children in different grades? And how?
2. Are there different types of storytelling activities? What are the processes of the storytelling activity like?
3. What are the effective or frequently used training methods?
4. Do children often encounter any problems? What are the problems? What scaffolding can be offered to children?
5. What digital devices/equipment does the school have? What are teachers’ and children’s digital literacy? Can children at various grades use computers on their own?

### B FOCUS GROUP PROTOCOL WITH CHILDREN

- What types of books do you like? What are some books that you have recently read? Could you please provide any examples?
- Do you engage in storytelling activities with your parents after school? If yes, when and how?
- If you were asked to create a story, what kinds of characters would you like to choose as the main character for your story? Could you give us 3-5 examples?
- What kinds of scenes would you like your story to take place in? Could you give us 3-5 examples?
- Have you ever used voice interaction in any digital devices (e.g., smartwatch)? Have you ever played computer games/software? Can you use a computer mouse and keyboard?

## C OTHERS

Children’s demographic details in the preliminary evaluation and their generated stories and comics can be

accessed from [https://drive.google.com/file/d/1LAu04X-GethZ8MbAeO\\_59vq46Y8r1tGc/view?usp=sharing](https://drive.google.com/file/d/1LAu04X-GethZ8MbAeO_59vq46Y8r1tGc/view?usp=sharing)